Sahaj Bhadja

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Summary

Software Engineer with 2+ years building and supporting production systems in **C#/.NET**, Python, and **SQL**, troubleshooting live customer issues, validating releases through simulation and testing, and explaining technical work to client stakeholders.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Software Engineer *Los Angeles, CA*

- Maintained integration jobs in live operations across **SQL Server**, PostgreSQL, and cloud storage, troubleshooting failures with reconciliation checks and error handling to sustain 99%+ reliability over 5K+ monthly jobs.
- Tuned **SQL** indexes and query patterns for a 500+ user platform, cutting dashboard latency from 1.2s to 350ms, and isolated failing background jobs to triple throughput across 1K+ jobs.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a motion planner for autonomous vehicles and validated it through **closed-loop simulation**, processing 100K+ trajectory logs and improving long-horizon trajectory planning 18% across simulated driving scenarios.
- Ran controlled **simulation tests** comparing a map-aware encoder against three baselines, documenting failure modes such as off-road trajectories and improving closed-loop planning stability 15% for safer behavior.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Automated release validation that built executables, installed them on Windows, Linux, and macOS test VMs, and ran 20+ **integration and unit tests** nightly with CSV reports and failure logs.
- **Troubleshot** large-scale API integrations from logs and applied object-oriented Factory and Singleton patterns, improving response times 30% and cutting technical debt 20% while working directly with Microsoft stakeholders.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built **C# .NET** services on Azure SQL for real-time resource allocation and assignment conflict resolution across multiple sites, syncing 2M+ records and reducing resource wastage 42% through analytics.
- Diagnosed production errors with Sentry, Azure Monitor, and Kibana and shipped fixes through Jenkins CI/CD with automated tests and approval-gated **staged releases** to production, cutting deployment time 50%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Worked directly with the client to map 12+ integration points and **presented the end-to-end architecture** for approval, then built Flask booking APIs with responses under 200ms across 50K+ records.
- Prevented double-booked time slots with **transactional locking** and automated refund and dispute handling through idempotent webhooks, cutting manual payment work about 70% across mobile and web clients.

Skills

Languages: C#, SQL, Python, Java, C++, JavaScript, Bash
Platforms: .NET, SQL Server, Azure SQL, PostgreSQL, MySQL, REST APIs, Microservices, Windows, Linux
Testing & Simulation: Simulation, Integration Testing, Unit Testing, PyTest, Mockito, Release Validation, Debugging
Support & Ops: Troubleshooting, Root Cause Analysis, Log Analysis, Sentry, Azure Monitor, Kibana, Technical Documentation
Tools & Practices: Git, Jenkins, CI/CD, Docker, Azure, AWS, Agile, Customer Communication, GitHub Actions, Postman

Projects

Warehouse Mission Simulator | *C#, .NET, SQL Server, xUnit*

- Designed a **C#** simulator that replays pallet transport missions for automated guided vehicles across a warehouse layout, detects path conflicts and blocked aisles, and validates **software upgrades** before site deployment.

Site Diagnostics Toolkit | *C#, SQL Server, PowerShell, Windows Services*

- Designed a field **diagnostics** toolkit that collects logs and database health from a customer site, flags stuck missions and slow SQL queries, and generates a support report for **remote troubleshooting**.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2, OS Scheduling*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel queue **scheduling simulations** in C++ with Gantt visualization, testing 1K+ process combinations and cutting average turnaround time 25%.